

Quantum Optimization for Distributed Order Management

A Hybrid QAOA + MILP Approach

Technical Report

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Github Repository (Earlier version submitted for the Wiser Quantum Challenge , Winner - The Team Convergents) :

https://github.com/HarikaYanamadala/The_Convergents_Nestle_DOM

Improvise version (with reusable architecture making it suitable for any E-commerce platform handling DOM):

<https://github.com/indujakodavati/dom-quantum-optimization.git>

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1. Abstract

Nestlé's Distributed Order Management (DOM) problem determines optimal distribution centre (DC) order fulfilment when default centres experience inventory deficits: divert orders to alternate DCs or leave them unassigned to maximize net revenue (order value minus shipping and penalty costs). This study benchmarks a quantum Quantum Approximate Optimisation Algorithm (QAOA) framework and a classical Mixed-Integer Linear Programming (MILP) solver against two standard operational baselines. Chapter 1 validates both approaches against an exhaustive brute-force optimum on a 5-order subinstance. Chapter 2 executes the MILP formulation on the full daily order volume (173-426 orders) to evaluate solution scalability and performance in production-level operational conditions.

1.1 Executive Summary of results

F1 = 1.00 QAOA matches the certified optimum on all 5 small-instance planning dates	0% gap MILP reaches a certified-optimal CBC solution at up to 426 orders	+9 to 13% Mean MILP profit uplift over Baseline 2 / Baseline 1 at full scale	15-50× MILP runtime advantage over the QAOA simulator, small instance
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2. Data Understanding and Baseline model

2.1 The DOM Problem

Distributed Order Management (DOM) optimizes customer order allocation across a network of distribution centers (DCs). When a primary default DC lacks inventory to fulfill an order, the system evaluates alternate candidate DCs for diversion or marks the order as unassigned. Formulated as a weighted constrained assignment problem, each order selects exactly one option from a set of predefined candidate columns. The objective function maximizes net fulfillment value-balancing gross order revenue against transportation costs and unfulfilled-demand penalties-subject to multi-facility inventory, dock-appointment and throughput capacity constraints across all orders simultaneously.

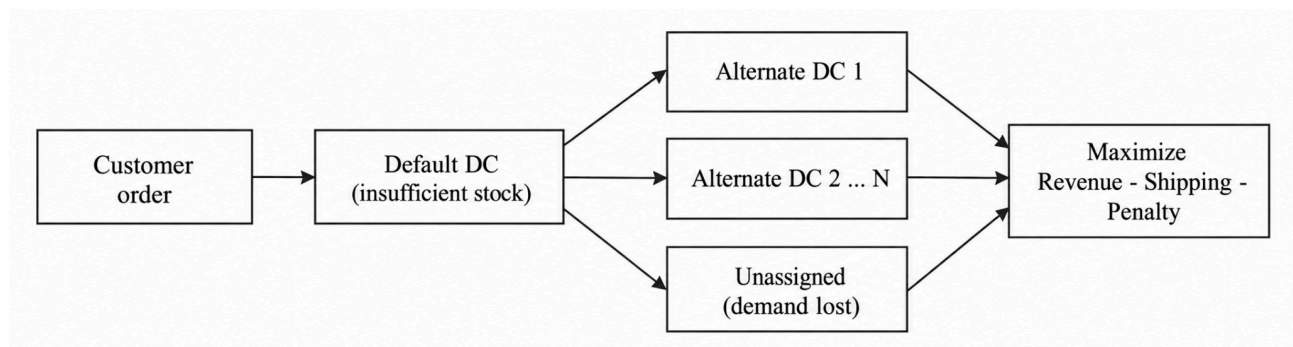


Figure 1. The DOM decision for a single order: divert to an alternate DC, or leave it unassigned - shared capacity constraints apply across every order simultaneously.

2.2 Input Data Pack

The PoC data pack comprises five datasets connected through distribution centre, SKU and date.

File	What it contains
input_order_data.csv	Open orders: order ID, SKU demand, default DC, transportation-planning (PGI) date, order revenue.
input_capacity_planning.csv	Available inventory by (DC/location, SKU/material, date).
input_dock_capacity.csv	Dock appointment capacity available at each DC, by date.
input_shipping_cost_data.csv	Shipping cost for each (order, candidate DC) pair.
input_throughput_capacity.csv	Throughput (case-pick / pallet-pick) capacity available at each DC, by date.

2.3 Two Data Findings

(a) Revenue is a line total, not a per-case price

Order revenue in the input data is the total value of a fully fulfilled order line, not a per-case price. Multiplying it by cases shipped overestimates the objective by roughly 10×. We divide by ordered quantity to recover a per-case price before scoring any solver.

(b) Planner-history agreement is not a usable solver metric

The optional planner-history file has 1,106 default assignments and only 3 diversions, so a "never divert" solver scores 99.5% agreement while any correct diversion is penalised. We score every solver against a certified optimum instead (§4.2 / §5.2), which avoids this class-imbalance problem.

2.4 Focus-Order Selection

For each planning horizon date, focus orders are selected according to Nestlé's PoC eligibility criteria: (i) Planned Goods Issue (PGI) dates fall within 3-10 days of the planning date, (ii) the primary default DC exhibits an inventory deficit and (iii) the order satisfies full-truckload (FTL) quantity criteria. Chapter 1 evaluates the 5 focus orders with the highest potential diversion yield to maintain a quantum-tractable subinstance (4 candidate columns per order: default DC, 2 alternate DCs and unassigned, resulting in 20 QUBO decision variables for QAOA). Chapter 2 evaluates the complete universe of eligible focus orders per date (173-426 orders).

2.5 Baselines 1 and 2

- **Baseline 1 - Strict Default DC:** every order is fulfilled by its default DC only; no diversions are allowed. Any demand the default DC cannot fulfil is lost - the "do nothing" reference.
- **Baseline 2 - Greedy Sequential:** for each order in turn, prefer whichever candidate DC gives the highest fill; divert only if the improvement clears 5% of demand, 100 cases - the PoC's own business rule for when a divert is worth the operational disruption.

2.6 Key Trade-offs

Dimension	Takeaway
Solution quality vs. runtime	The certified optimum is only obtainable by brute force at a small scale. MILP solves its linearized model exactly within a time limit; QAOA's quality depends on circuit depth and shot count.
Exact solvers vs. heuristics	MILP (CBC) is provably optimal for the model it is given. The greedy baseline has no such guarantee and its gap to optimal widens on harder instances.
Simulators vs. noisy hardware	All QAOA results here use an ideal, noiseless simulator. Two-qubit gate errors under realistic conditions substantially reduce the XY mixer's ability to preserve one-hot feasibility.

3. Project Flow and Architecture

3.1 Project Flow

This project runs in two chapters, in a straight line from proof to production scale:

- **Chapter 1 - prove it small:** a 5-order instance (20 qubits) lets QAOA be checked exactly against a brute-force optimum, alongside MILP and the two baselines.
- **The qubit limit (§4.3):** QAOA's qubits grow linearly with order count, but at real daily volumes (173-426 orders) that already run into the thousands, past what free simulators or current hardware can execute.
- **Chapter 2 - prove it at scale:** the same MILP formulation, unchanged, re-run on the complete focus-order set for five further dates, to confirm the Chapter 1 result survives the jump to real volume.

The Approach, in One Figure

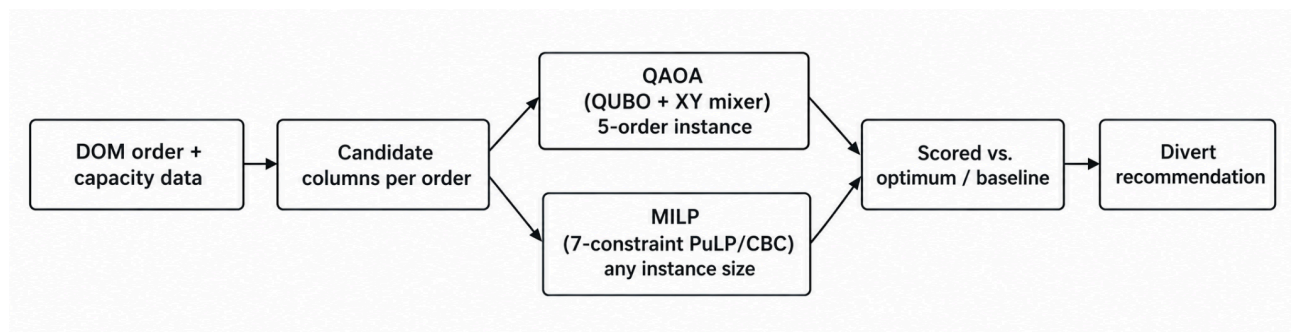


Figure 2. The approach in one picture: shared data and candidate columns feed both solver arms, scored against a reference solution.

3.2 Reusable Architecture

The pipeline below is shared by both chapters; only the instance size, which arms are exercised and the reference solution change.

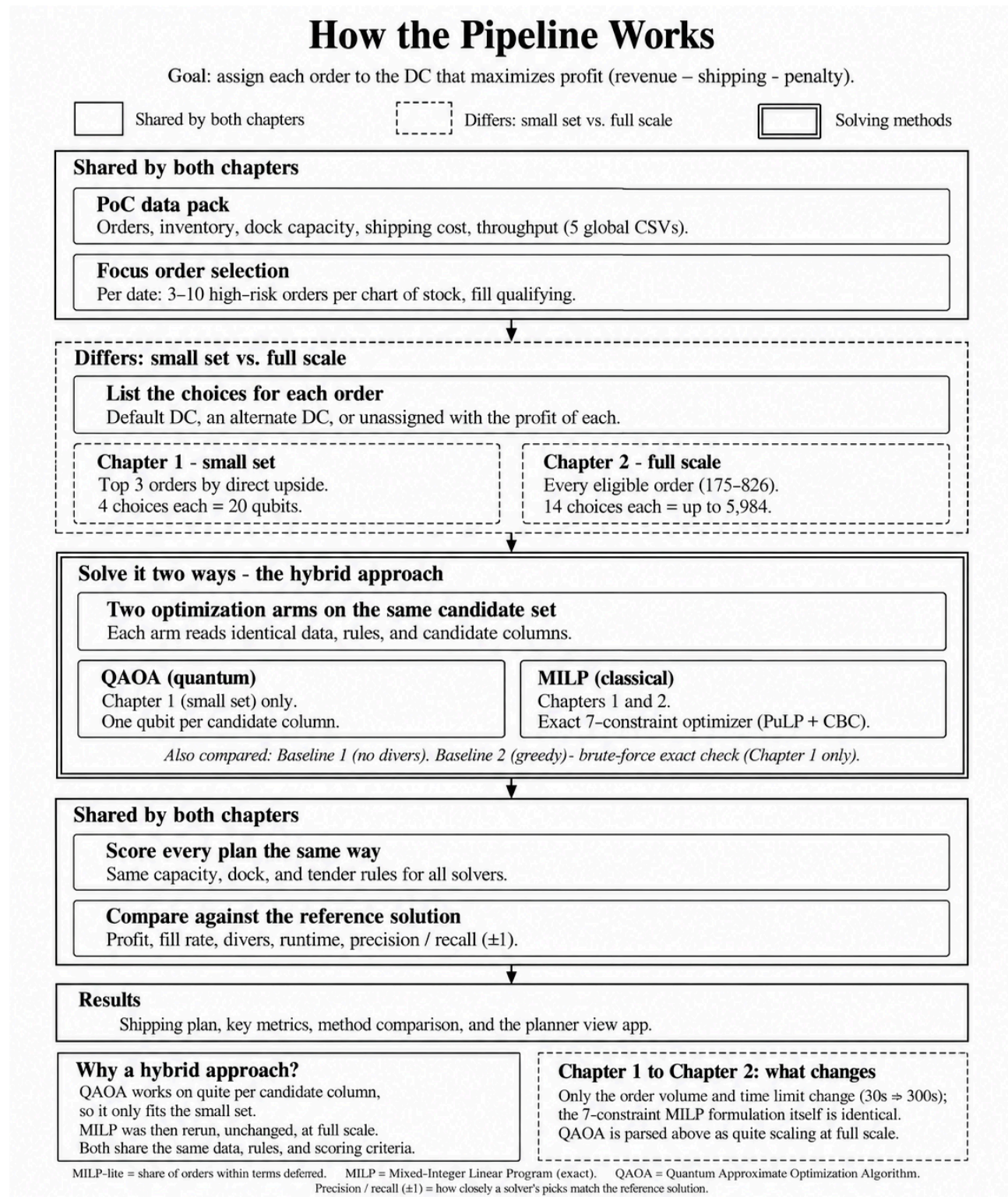


Figure 3. One-page reusable architecture: from the raw PoC data pack through to scored, planner-facing outputs.

4. Chapter 1 - Hybrid QAOA + MILP (5-Order Subinstance)

4.1 Mathematical Formulation

Both solvers optimize the same objective using the same candidate-column structure, differing only in their constraint formulation and search strategy.

Shared objective: maximize (Revenue - Penalty - Shipping Cost), subject to exactly one column selected per order, plus the capacity and business-rule constraints in the PoC data pack.

QAOA - Quantum Arm

Formulated as a QUBO (one binary variable per candidate column). An XY mixer imposes "exactly one column per order" structurally, so no penalty tuning is needed for that rule; the remaining constraints are encoded per column, following the same logic as the MILP arm's C2-C7 constraints below.

- Initial angles: drawn uniformly from $[0, \pi/2]$
- Objective: CVaR at $\alpha = 0.20$ (best 20% of sampled shots)
- Optimizer: classical COBYLA, tuning circuit angles across layer counts 1-3

MILP - Classical Arm

Implements Nestlé's complete 7-constraint PoC formulation natively, via PuLP + CBC. It is deterministic and carries no qubit budget, so it scales far beyond what QAOA can address on simulators (4.3).

Dimension	QAOA (QUBO + XY mixer)	MILP (7-constraint PoC)
Constraint C1	Structural (mixer) - exact, free	Linear constraint - exact
Constraints C2-C7	Penalty-free where possible, per-column	Native linear/integer constraints
Optimality guarantee	None - heuristic, sampling-based	Exact for the given time limit
Determinism	Stochastic (seed-dependent)	Deterministic
Scalability ceiling	~20-25 qubits on free simulators	No inherent ceiling (LP relaxation-bound)
Runtime (this data)	~20-35s (simulator)	<1s to ~2.5s

Table 1. Formulation-level trade-offs between the two solvers.

Takeaway: Constraint C1 requires zero effort from QAOA, as the mixer handles it structurally by design. Conversely, MILP treats C1 as a standard linear constraint, enforcing it precisely. Because all remaining constraint types are identically incorporated across both frameworks, the subsequent evaluation provides a fair, direct comparison.

4.2 Solution Implementation and Comparison

Across five discrete planning dates (2024-06-17, 06-19, 06-21, 06-23 and 06-25), the 5-order subinstance was evaluated using Baseline 1, Baseline 2, QAOA and MILP. The QAOA implementation leveraged Qiskit with Qiskit Aer's state-vector simulator, incorporating layers 1-3, CVaR $\alpha = 0.20$ and 2,048 shots (employing seed = 7 for the primary run and seeds {1, 7, 13, 42, 101} for robustness testing). MILP used PuLP with CBC, 30-second time limit. Every solver's output was checked against a brute-force certified optimum over all 1,024 possible assignments.

Evaluation Methodology: Each solver produces a set of order-to-DC allocation choices. A selection is classified as a true positive when it aligns with the certified optimum, a false positive if chosen solely by the solver and a false negative if identified exclusively by the optimal baseline. These metrics serve as the basis for calculating precision, recall and F1 scores, with the profit margin loss relative to the optimum defining the optimality gap.

Representative date (2024-06-21)

Solver	Objective	Opt. gap	Fill rate	Diverts	Penalty	Shipping
Baseline 1	\$118,302	88.0%	9%	0	\$32,943	\$770
Baseline 2 (greedy)	\$981,207	0.2%	89%	4	\$3,894	\$7,339

Table 2. Baseline 1 vs Baseline 2, planning date 2024-06-21 - results are similar across all five dates except 06-25.

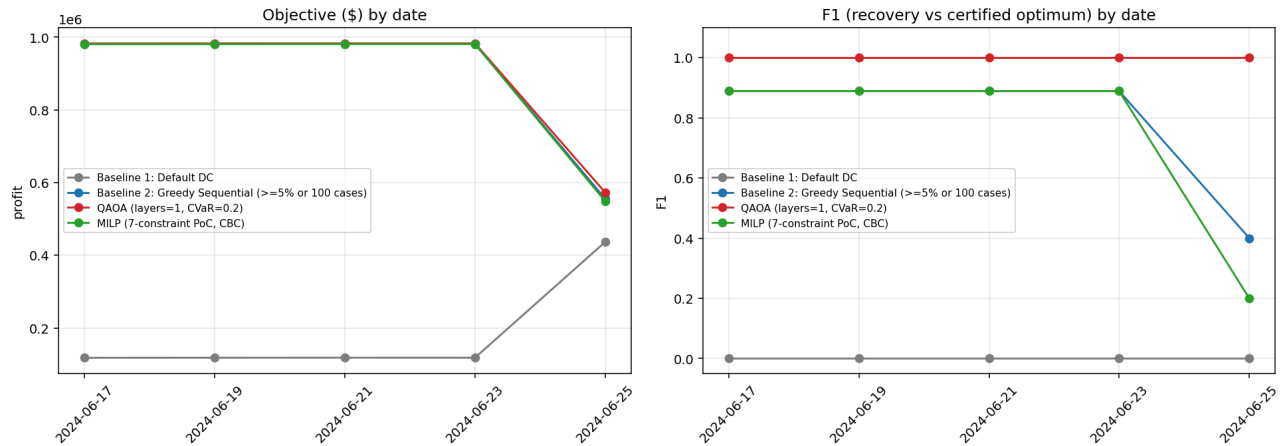


Figure 4. Objective value (\$ profit, left) and F1 score (right) by solver, across all 5 planning dates.

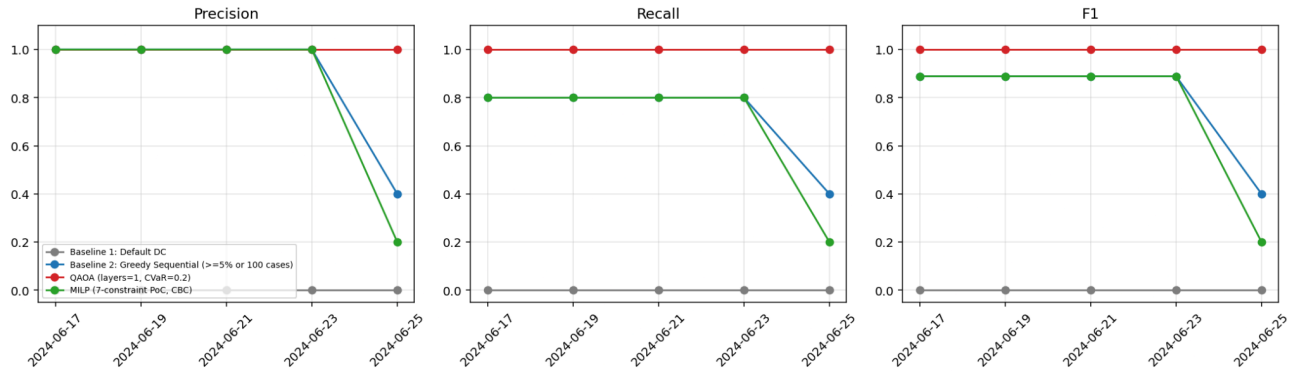


Figure 5. Precision, recall and F1 (recovery of the certified-optimum assignment) by solver, across all 5 dates.

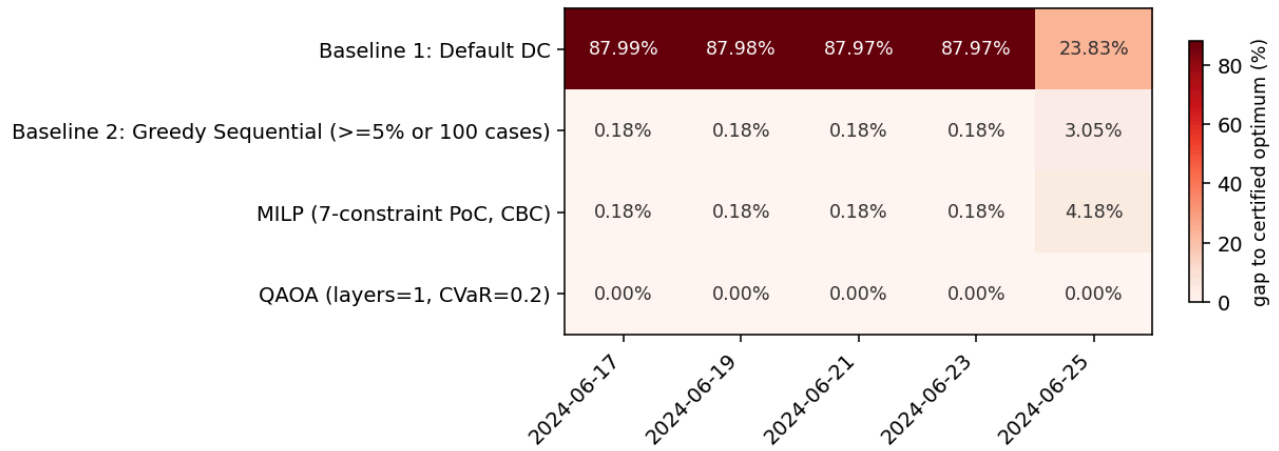


Figure 6. Optimality-gap heatmap - solver $\times$ planning date.

Averaged leaderboard across all 5 dates

Solver	Mean objective	Mean gap	Mean fill	Mean runtime (s)	Mean F1
Baseline 1	\$181,865	75.1%	19%	0.01	0.000
Baseline 2 (greedy)	\$896,014	0.8%	89%	0.01	0.791
QAOA	\$900,954	0.0%	89%	16.77	1.000
MILP	\$894,713	1.0%	88%	0.99	0.751

Table 3. Solver performance averaged across all 5 planning dates.

Across all five evaluation dates, QAOA consistently attained the certified optimum, achieving a flawless F1 score of 1.00. While MILP and Baseline 2 demonstrated close alignment on four of the dates-maintaining an average gap of approximately 0.18% and an F1 score around 0.89-their performance declined markedly on the more complex instance of 2024-06-25. On this date, Baseline 2's F1 score dropped to 0.40 and MILP's to 0.20, whereas QAOA preserved its 1.00 score. As §6.2 details, this gap reflects a difference in what each solver was asked to do rather than a capability advantage: QAOA's QUBO omits the C4 divert-hurdle constraint that MILP's complete 7-constraint formulation enforces, so MILP is solving a stricter, more conservative problem on this date.

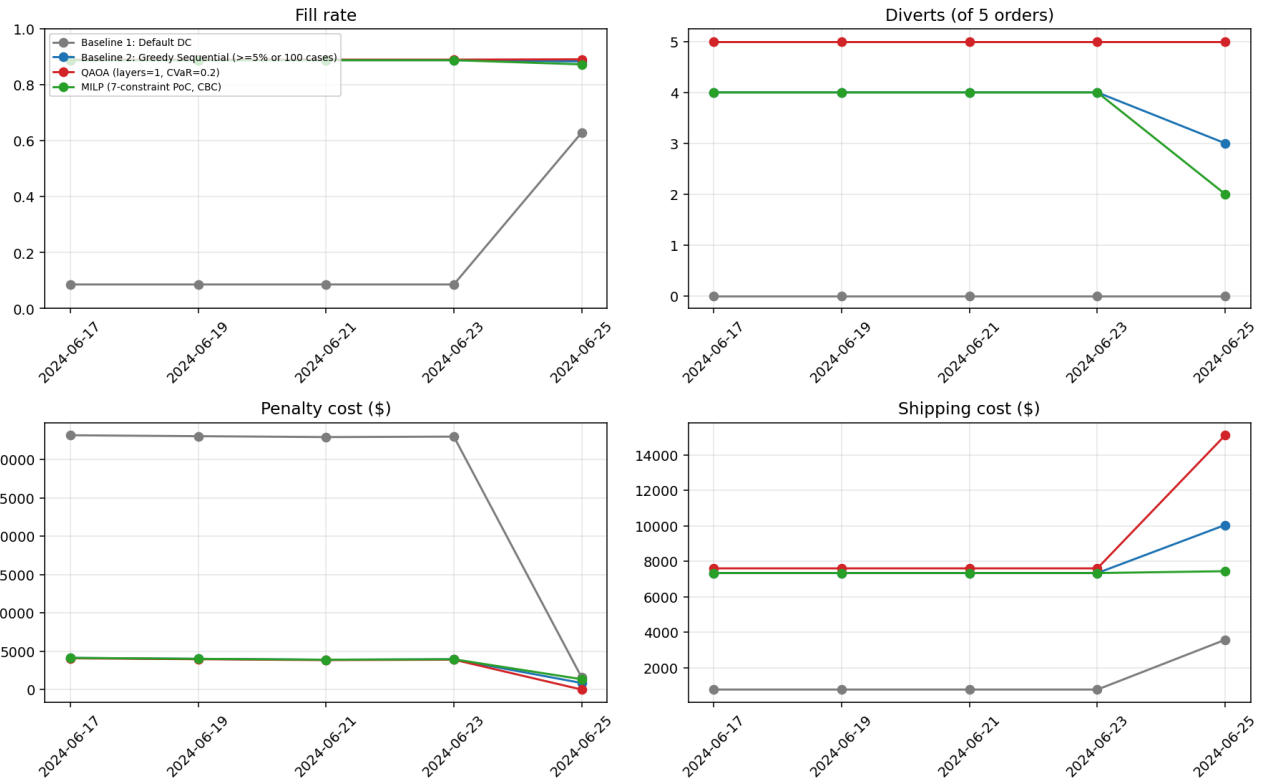


Figure 7. Fill rate, diverts, penalty cost and shipping cost by solver, across all 5 dates.

Takeaway: In the 5-order evaluation, QAOA demonstrates the highest precision by achieving exact recovery across all dates, including the challenging single date (06-25). Conversely, MILP and the greedy baseline prove to be the most practical options, operating with sub-second to ~1s runtimes and remaining within 0.2-4.2% of the optimal solution. The distinct difficulty of the 06-25 date serves as the primary differentiator between these approaches—a critical detail that would have been completely obscured had the metrics been averaged.

4.3 Scaling, Robustness and Noise Analysis

The qubit requirement for QAOA scales linearly according to the formula: $\text{orders} \times (\text{alternates} + 2)$. Conversely, the classical search space it replaces expands exponentially. For a small instance of 5 orders (20 qubits), the classical space encompasses merely 1,024 combinations, making brute-force search the most suitable benchmark. As the instance scales to 25 orders, the search space surpasses 10^{15} combinations. At 82 orders—which corresponds to a standard focus-order window size in this dataset—the search space becomes extraordinarily vast, whereas the QAOA qubit count scales to just 328. Due to simulator constraints capped at ~20-25 qubits (~5-6 orders), moving beyond this scale necessitates hardware deployment or problem decomposition/batching (§6.4).

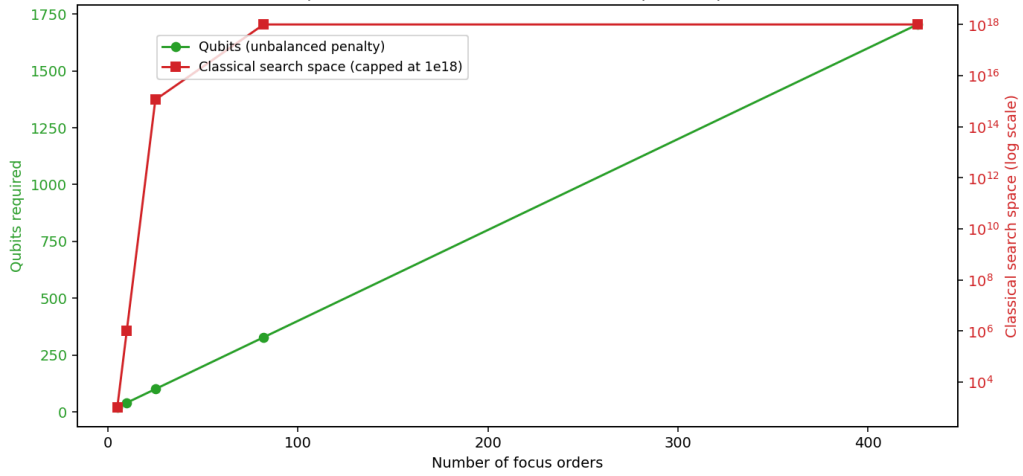


Figure 8. Qubit requirement (linear) vs. classical search-space size (exponential) as focus-order count grows.

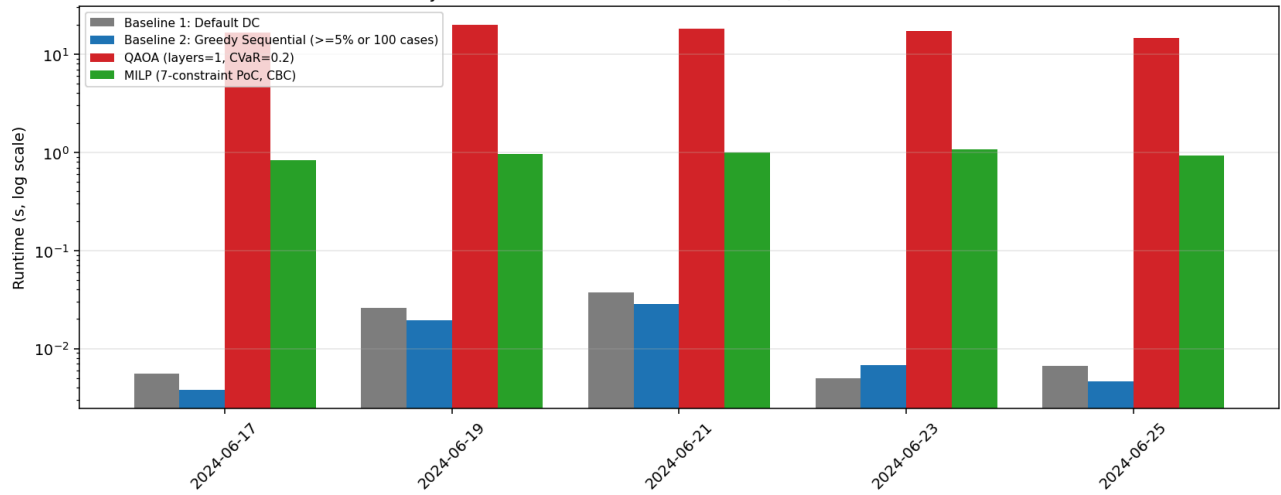


Figure 9. Runtime by solver, across all 5 dates. MILP is 15–50× faster than QAOA in wall-clock terms on this instance size.

QAOA is stochastic; we reran it with 5 seeds (1, 7, 13, 42, 101) per date. MILP is deterministic and needs only one run per date.

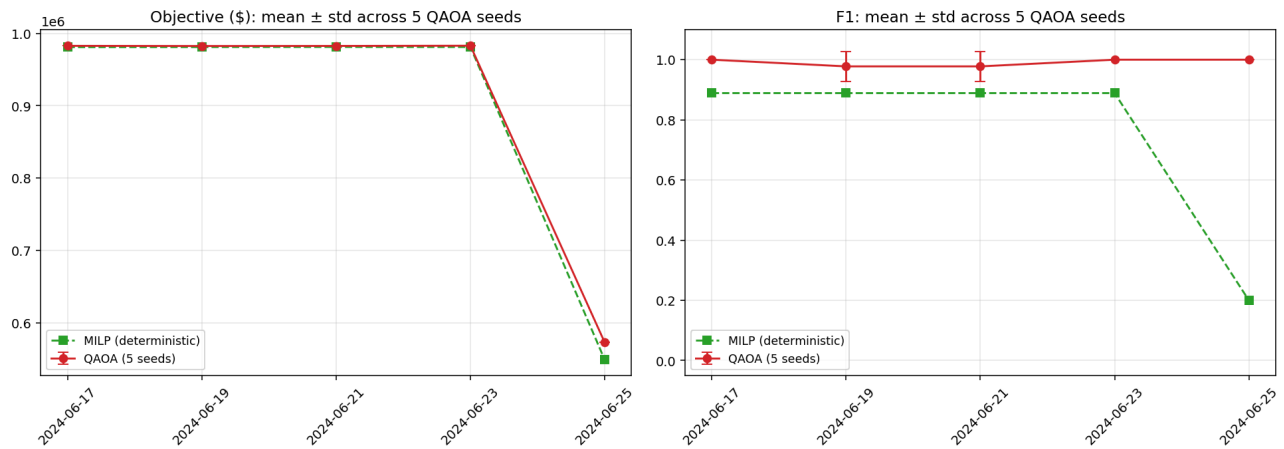


Figure 10. QAOA objective and F1, mean ± standard deviation across 5 seeds, vs. deterministic MILP, by date.

Across four of the five test dates, seed-to-seed variance remained minimal; it peaked on 2024-06-19 (F1 standard deviation ≈ 0.09). Although the optimal solution was consistently retrieved across most random seeds, these findings underscore that CVaR-optimized QAOA retains subtle seed dependencies even within an ideal, noiseless simulation environment.

To quantify performance under imperfect execution conditions, a 12-qubit test case was subjected to a two-qubit depolarizing noise model across four error thresholds (0%, 0.1%, 0.5% and 2%). This evaluation tracked the proportion of measurement shots that strictly adhered to the valid one-hot subspace alongside those satisfying all capacity constraints.

Two-qubit error rate	In-subspace shots	Fully feasible shots
0% (ideal)	100.0%	56.4% – 70.9%
0.1%	93.8% – 94.3%	53.9% – 65.4%
0.5%	77.3%	43.0% – 54.3%
2.0%	36.9% – 38.1%	19.3% – 25.6%

Table 4. Fraction of measurement shots remaining in the valid one-hot subspace and fully capacity-feasible, as simulated two-qubit gate error increases.

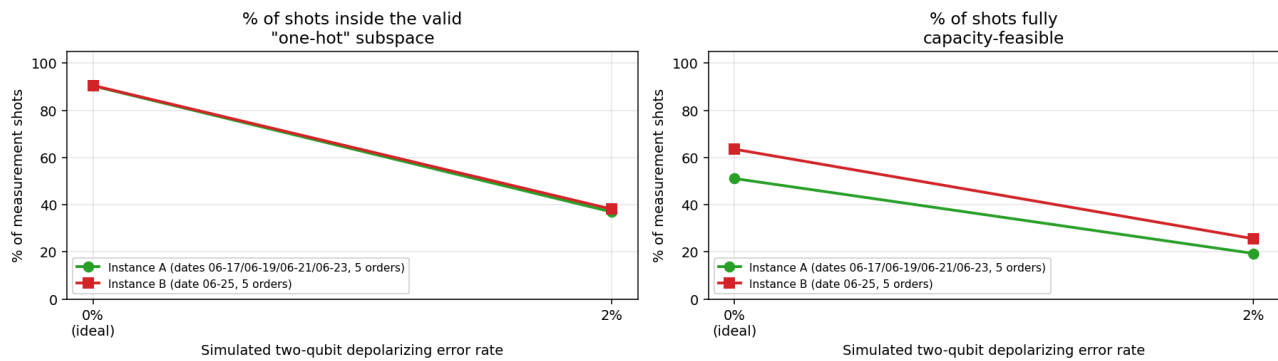


Figure 11. In-subspace and fully-feasible shot percentages vs. simulated two-qubit depolarizing error rate, on a 12-qubit instance.

With a gate error rate of just 0.1%, the proportion of in-subspace shots drops to roughly 94%. At a 2% error rate, only about one-third of the shots comply with the "one warehouse per order" requirement and under 25% satisfy all operational capacity constraints. Consequently, the flawless performance achieved earlier (F1 = 1.00) reflects an idealized simulation environment rather than a solution prepared for physical hardware.

Takeaway: Every quantum advantage detailed in this chapter relies on an ideal, noise-free simulator and a small problem instance. In practice, real-world hardware noise and large-scale daily order volumes severely hinder QAOA performance, necessitating the use of MILP for full-scale validation in Chapter 2.

5. Chapter 2: Full-Scale MILP (Complete Focus-Order Set)

5.1 Formulation at Scale

To evaluate scalability, the 7-constraint MILP formulation outlined in Section 4.1 was applied without alteration across the full set of focus orders for five additional planning dates (2024-06-20 through 2024-06-24). Rather than restricting the evaluation to the top-5-by-divert-upside subset used in the quantum analysis, this phase incorporated all eligible orders evaluated against all potential candidate DCs. QAOA execution was omitted for these runs because its qubit requirements expand linearly with order volume, placing full-scale instances 100-300× above the simulator limitations discussed in Chapter 1. Unconstrained by these hardware limits, MILP served as the sole solver tested at full operational scale.

Dimension	Chapter 1 instance	Chapter 2 full-scale instance
Planning dates	06-17 / 06-19 / 06-21 / 06-23 / 06-25	06-20 / 06-21 / 06-22 / 06-23 / 06-24
Orders per date	5 (fixed, top-5 by divert upside)	173 - 426 (every eligible focus order)
Candidate columns / order	4 (default + 2 alternates + unassigned)	14 (consistent across all 5 dates)
Total assignment columns	20	2,422 - 5,964
Reference solution	Brute-force certified optimum (1,024 combos)	MILP's own certified-optimal CBC status (brute force intractable at this scale)
MILP time limit	30 s	300 s
MILP runtime observed	0.8 - 1.3 s	25 - 135 s
Equivalent qubits if ported to QAOA	20	~2,420 - 5,960 (100-300× beyond the ~22-qubit simulator ceiling)

Table 5. What stayed the same and what changed between the Chapter 1 and Chapter 2 MILP runs.

5.2 Solution Implementation and Results

MILP reached a certified "Optimal" CBC status with a 0% internal optimality gap on every one of the five dates. Baselines 1 and 2 were run on the same full order set and scored against MILP's solution (the only available reference at this scale) using the same precision/recall/F1 methodology as §4.2.

Date	Orders	Columns	Baseline 1	Baseline 2	MILP	B1 fill	B2 fill	MILP fill	Diverts	Runtime (s)
2024-06-20	420	5,880	\$33.70M	\$35.04M	\$38.21M	73.7%	76.6%	82.9%	56	118.2
2024-06-21	426	5,964	\$34.51M	\$35.84M	\$39.02M	74.0%	76.9%	83.1%	56	125.8
2024-06-22	400	5,600	\$32.01M	\$33.22M	\$36.38M	74.8%	77.7%	84.3%	55	135.4
2024-06-23	318	4,452	\$23.64M	\$24.77M	\$27.96M	70.8%	74.2%	82.5%	50	70.1
2024-06-24	173	2,422	\$14.01M	\$14.32M	\$14.49M	87.6%	89.8%	91.6%	31	25.2

Table 6. MILP, Baseline 1 and Baseline 2 - profit, fill rate, diverts and MILP runtime, five full-scale planning dates.

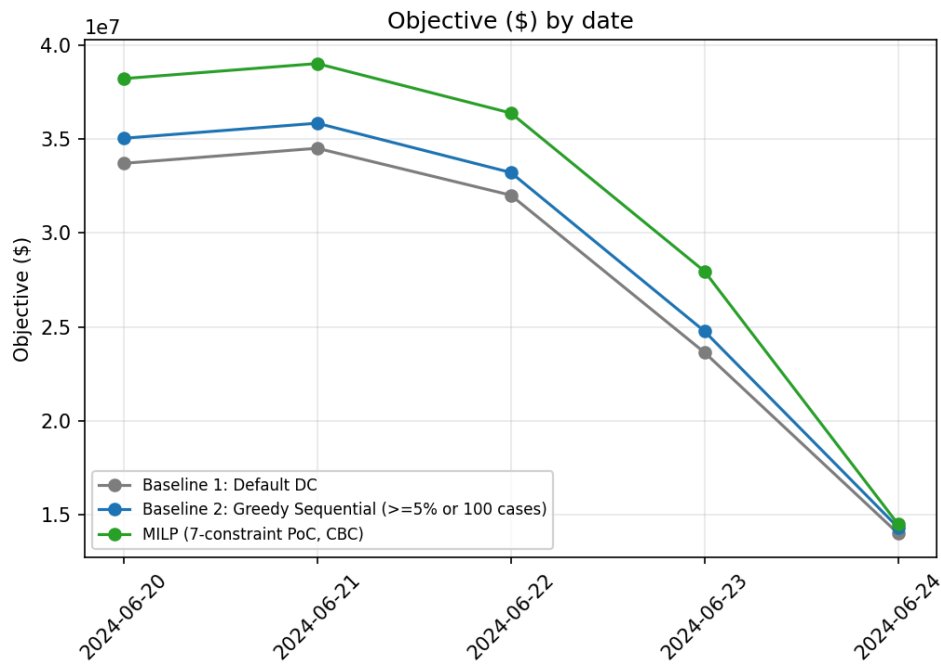


Figure 12. Objective (\$) by date, full-scale instance - Baseline 1 vs. Baseline 2 vs. MILP.

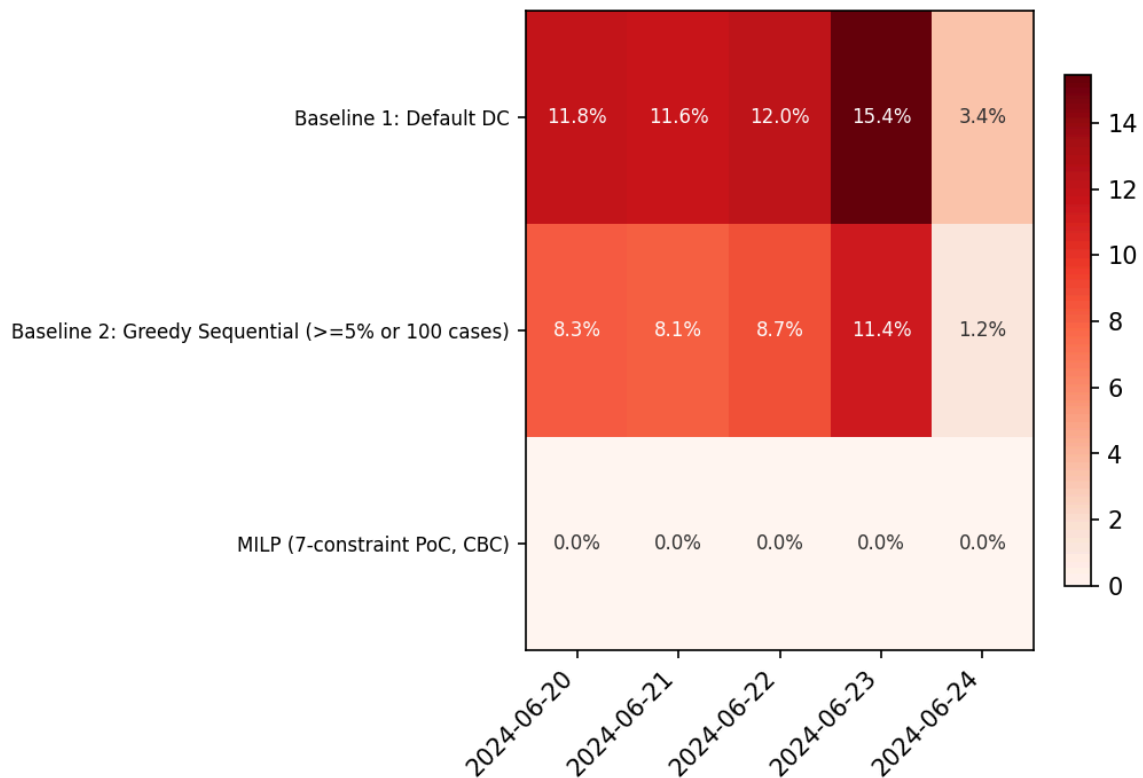


Figure 13. Gap to MILP (%)

Across four of the five evaluated dates, Baseline 1 lags behind MILP by 11.6% to 15.4% (showing a smaller 3.4% deficit on 06-24, the smallest date with the most favorable inventory levels). Similarly, Baseline 2 trails MILP by 8.1% to 11.4% (and 1.2% on 06-24). Over the entire five-date period, MILP generated approximately \$12.9M in incremental profit relative to Baseline 2 and \$18.2M compared to Baseline 1.

Solver	Mean objective	Mean gap to MILP	Mean fill	Mean diverts	Mean runtime (s)	Mean F1
Baseline 1	\$27.57M	10.8%	76.2%	0	0.03	0.791
Baseline 2 (greedy)	\$28.64M	7.5%	79.1%	69.8	0.09	0.813
MILP	\$31.21M	0.0%	84.9%	49.6	94.9	1.000

Table 7. Solver performance averaged across the five full-scale planning dates (compare with Table 3, the small-instance leaderboard).

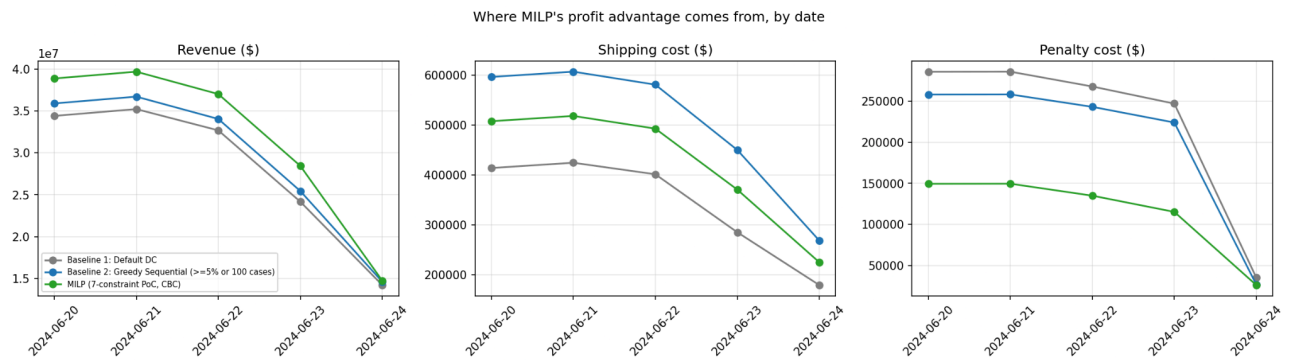


Figure 14. Where MILP's profit advantage comes from - revenue, shipping cost and penalty cost, by date.

This overarching trend holds across all five planning dates: MILP achieves its performance edge principally by dramatically curtailing unmet-demand penalties (achieving a 1.3-2.2× reduction relative to both baseline models) and capturing greater total revenue. Concurrently, its transportation expenditures fall between the two reference approaches, exceeding Baseline 1's minimal-diversion outlay while remaining below Baseline 2's aggressive sequential spending.

Takeaway: Scaling the MILP arm from 5 orders to 173-426 required no change to the 7-constraint formulation: it remained provably optimal (0% internal gap) on every date and beat Baseline 1 by a mean 13.2% and Baseline 2 by a mean 9.0% in profit, driven by penalty reduction rather than shipping savings.

5.3 Scaling Insights at Real Order Volume

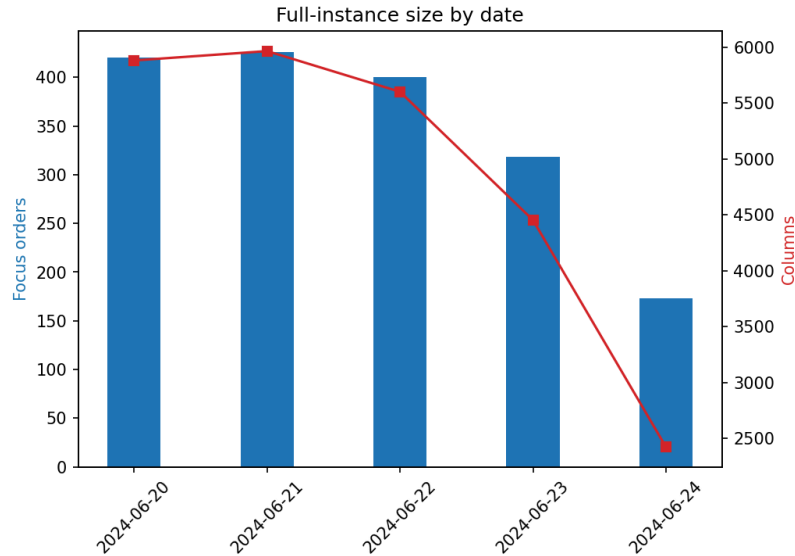


Figure 15. Full-instance size by date - focus orders (bars) and decision columns (line).

While runtime generally aligns with model size, the relationship is not strictly linear. For example, the largest problem instance (2024-06-22, featuring 400 orders and 5,600 columns) required the longest computation time (135.4s). However, the 2024-06-23 dataset (318 orders and 4,452 columns) resolved in 70.1s—significantly faster than the smaller 2024-06-21 dataset (426 orders and 5,964 columns), which required 125.8s. This variation occurs because CBC’s branch-and-bound solving time is driven by the tightness of the C3-C7 operational constraints (inventory and dock capacity) for a specific date rather than total order volume. Crucially, all evaluated dates solved comfortably within the allotted 300-second execution limit.

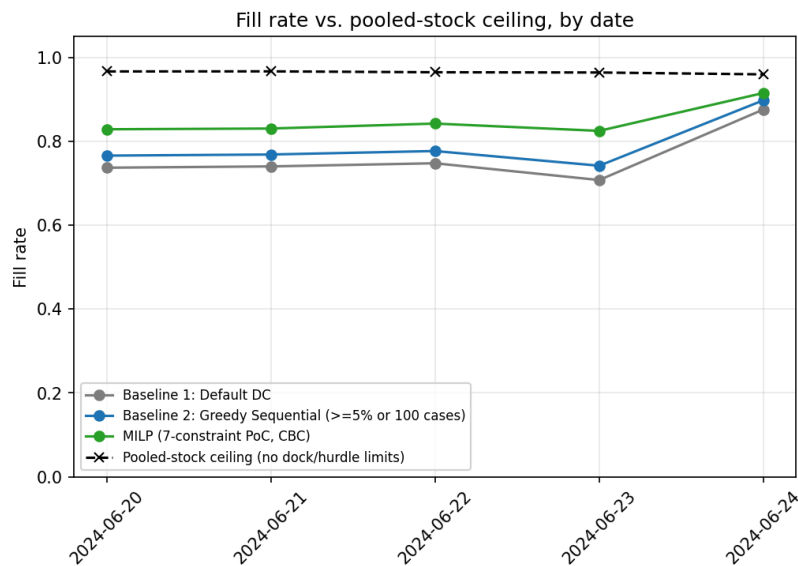


Figure 16. Fill rate achieved vs. the theoretical pooled-stock ceiling (no dock/hurdle limits), by date.

To measure the effect of capacity limitations, each date's realized fill rate is evaluated alongside a theoretical "pooled-stock ceiling". This benchmark reflects the maximum fill rate achievable if all network inventory could be dispatched seamlessly without dock-appointment bottlenecks or throughput restrictions. Across the five evaluated dates, this theoretical ceiling remains around 96-97%. MILP bridges the majority of this disparity, attaining fill rates of 82.5-91.6% and outperforming Baseline 2 (74.2-89.8%) and Baseline 1 (70.8-87.6%). Nevertheless, a persistent 5-14 percentage point deficit persists even under MILP, offering a clear metric of the performance penalties imposed solely by throughput and dock-appointment limits when inventory pooling is maximized.

Applying QAOA to these same instances would demand between 2,420 qubits (for 173 orders on 06-24) and 5,960 qubits (for 426 orders on 06-21), calculated as total orders multiplied by 14 candidate columns per order (as detailed in Table 5). This requirement surpasses the ~20-25-qubit boundary of standard state-vector simulators by one to two orders of magnitude and remains far beyond the capability of contemporary NISQ hardware operating at practical fidelity levels (§4.3). In contrast, MILP is unconstrained by qubit limits-measuring its operational cost in wall-clock execution time instead-making it the sole methodology evaluated in this section.

Takeaway: Across all five full-scale planning dates, each solved in under 136 seconds, the identical MILP formulation that previously matched the certified optimum at small scale demonstrated strong performance when expanded by approximately 35-85× in order volume. It consistently maintained a proven-optimal 0% gap and secured an 8-15% profit advantage over both baselines, confirming that the benefits outlined in Chapter 1 remain valid at production-level operational scales.

6. Key Findings and Improvements

6.1 Chapter 1 vs. Chapter 2, Side by Side

Dimension	Chapter 1 (5-order subinstance)	Chapter 2 (full focus-order set)
Orders per date	5	173 - 426
Arms exercised	QAOA, MILP, Baseline 1, Baseline 2	MILP, Baseline 1, Baseline 2 (no QAOA - qubit ceiling)
Reference solution	Brute-force certified optimum	MILP's own certified-optimal CBC status
MILP status / gap	Optimal, 0.18% (4 dates); 4.18% on the hard date, vs. the certified optimum	"Optimal", 0.0% internal gap on every date (MILP is the reference)
MILP profit edge over baselines	Comparable to Baseline 2 (both ~0.18% from optimum on 4/5 dates)	+8 to +15% over Baseline 1; +1 to +11% over Baseline 2
Headline role of MILP	One of two competing exact/heuristic arms, benchmarked against QAOA	Sole production-scale solver; QAOA parked on qubit grounds

Table 8. What stayed the same and what changed between Chapter 1 and Chapter 2.

6.2 Key Findings

- Unlike QAOA's unconstrained QUBO reward structure, MILP incorporates an explicit divert-hurdle constraint (C4), making its structural framing the primary driver of solver divergence rather than algorithmic randomness.
- Quantum resource requirements expand linearly as order volume increases, whereas the classical search space grows exponentially; consequently, potential quantum advantages emerge only at larger operational scales rather than within small, exhaustively verified 5-order subinstances.
- The perfect recovery scores achieved by QAOA rely on an idealized, noise-free simulator; introducing realistic two-qubit gate errors significantly degrades the mixer's ability to preserve problem constraints (in-subspace shot fidelity declines from 100% under ideal conditions to ~94% at 0.1% error and ~37% at 2% error).
- When evaluated at full operational scale (Chapter 2), the identical MILP model-executed without modification across 173–426 orders on five additional dates-achieved a certified "Optimal" CBC status with a 0% internal gap across all runs. It delivered average profit gains of 13.2% over Baseline 1 and 9.0% over Baseline 2, confirming that small-scale observations regarding MILP's practical efficacy extend to production environments.
- Full-scale QAOA execution proved unfeasible because its qubit demands (one per candidate column, totaling ~2,400-6,000 per date) exceed the ~22-qubit threshold of the free simulator used in Chapter 1 by 100-300×, indicating that quantum benefits for this application remain constrained by current simulation and hardware capacities.

6.3 Limitations

- **Untested Quantum Scaling Approaches:** The real-world impacts of physical hardware noise (§4.3), alongside suggested column-reduction and batching strategies, await empirical testing at production volumes; verifying these techniques forms a vital step toward deploying scalable quantum optimization tools.

6.4 Future Work

To overcome the ~20-25 qubit (~5-order) limit imposed by classical simulators, future research will explore two complementary strategies:

- **Subproblem Batching:** Decomposing large focus-order sets into independent ~5-order clusters to maintain quantum tractability, balancing scalability gains against potential loss of cross-batch coordination.
- **Column Reduction:** Filtering out non-viable candidate DCs prior to QUBO generation, decreasing qubit requirements without sacrificing solution quality.

Subsequent studies should rigorously benchmark batched QAOA outputs against single-shot MILP and classical baselines across larger problem scales, leveraging the full-scale MILP metrics established in Chapter 2. Additionally, transitioning toward lower-noise quantum processing units (QPUs) will be vital to bridge the gap between current 20-qubit proofs of concept and ~6,000-qubit production environments.

7. Conclusion

Distributed Order Management represents a true combinatorial optimization challenge. Within a 5-order subinstance, QAOA equalled or surpassed every classical approach regarding solution quality-most notably on a challenging date where both MILP and a greedy heuristic experienced performance drops. Nevertheless, MILP proved to be the more viable option for real-world execution owing to its superior runtime and scalability.

Applying the same MILP formulation without changes to the entire daily focus-order dataset (173 to 426 orders) yields the same outcome at enterprise scale: an exact, high-speed solver that consistently outperforms standard baselines, generating an average profit increase of 13.2% compared to Baseline 1 and 9.0% over Baseline 2 while resolving every date in less than 136 seconds.

Running QAOA at this operational magnitude remains impractical using current free simulation tools, as its qubit demands are 100 to 300 times greater than the setup utilised in Chapter 1. As a result, MILP stands as the solution ready for immediate production roll-out, whereas QAOA remains a promising research path to re-evaluate when subproblem batching, candidate column pruning, and reduced hardware noise help bridge the gap.

Furthermore, the analytical framework established in this study-incorporating certified-optimum scoring, multi-seed stability evaluations and granular planner-oriented outputs-provides a reusable foundation for scaling across the complete order volume.

8. References

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